MAWROOTH - موروث

SAUDI IDENTITY TRACK

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ABSTRACT

Mawrooth is an AI-powered application that makes Saudi Nabati poetry more accessible to younger generations by explaining poetic verses in clear language and transforming their imagery into visual representations.

Nabati poetry contains distinctive vocabulary, cultural references, and poetic expressions that can be difficult for modern readers to understand. Mawrooth addresses this challenge through a fine-tuned ALLaM-7B model, adapted using QLoRA and supported by a specialized dictionary of Nabati terms to improve heritage-vocabulary understanding.

The system generates a structured explanation of each verse, which is then used to create a visual representation of the poetic scene through generative AI. By combining Arabic language processing with visual generation, Mawrooth provides an interactive way to explore Saudi literary heritage and make traditional poetry more accessible to new generations.

IDEA FIELDS

Mawrooth operates at the intersection of Natural Language Processing, Generative AI, and Digital Heritage Preservation. In the cultural and educational domain, it serves as an intelligent literary companion, helping students, researchers, and heritage enthusiasts understand classical Nabati poetry without specialized linguistic expertise.

In Arabic NLP, the project addresses a significant challenge: processing low-resource dialectal and archaic vocabulary that differs substantially from Modern Standard Arabic.

The generative visual component connects computational linguistics with digital art, creating a multimodal experience rarely explored in Arabic heritage technology.

The project also contributes to cultural tourism and national identity preservation, supporting institutions seeking to digitally document and promote Saudi literary heritage for both local and international audiences.

DATABASE
586 EXAMPLEA

DICTIONARY
+1000 TERMS

COT CONVERSION
5 SECTIONS

DATA SPLITTING
TRAIN 527 | EVAL 59

LORA FINE-TUNING
ALLAM-7B +QLORA-4BIT
R=16 ALPHA=32

EVALUATION
TOKEN ACCURACY :
85.75%

INTERFACE
BASE+LORA+
DICTIONARY

DEPLOYMENT
GRADIO INTERFACE



RESULTS

The fine-tuned Mawrooth model, based on ALLaM-7B and adapted using QLoRA, achieved a token-level accuracy of 85.75% on a held-out evaluation set of 59 examples after training on 527 examples.

The model generated structured Modern Standard Arabic explanations of Nabati verses, while the integrated 1,328-term dictionary supported the interpretation of heritage vocabulary during inference.

Qualitative testing also demonstrated the model's ability to generate coherent explanations for unseen verses.

FUTURE WORK

Future development includes:

- Expanding the dataset to cover more diverse Saudi dialects and regions.
- Further refining model training to improve generalization on unseen verses.

- Adding a feature to generate a complete heritage research article for each analyzed verse.

- Developing a full mobile application for broader accessibility to heritage enthusiasts and researchers.

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